CIS 4910

Computer Science & Engineering Project: Capstone

Requirements, Specifications The SRS Cornerstone

WEEK 3 • FALL 2026

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WEEK 3

Agenda for Today

1. **PRESENTATIONS**
2. An example of the Process
3. Review Epics
4. Review Features
5. Review about Business Requirements
6. Review Engineering Specifications
7. Requirement Levels (L1, L2, L3)
8. Producing the Specification
9. The complete End-2-End Process

SHORT PRESENTATIONS

Week 3

Random Order: Teams: 4, 9, 1, 2, 6, 8, 5, 3, 7

How Does the Process Work?

A Patent Example

EXAMPLE: PATENT AS A SPECIFICATION

Traffic Stop Sign Safety Enhancement System

Patent 2-241914 • Granted October 31, 1916 • Category: Traffic Control Devices

PATENT
Stop Sign Safety System

Size

Physical dimensions of sign face, post, and reflective surface.

Color

Retroreflective red with white border for low-light visibility.

Location

Right-side, left-side, or overhead mounting by intersection type.

Road Markings

Pavement markings in the approach zone warn drivers ahead.

Warning Distance

Minimum distance from warning sign to stop sign by speed limit.

Warning Sign

Upstream advisory sign alerts drivers before they see the stop.

EXAMPLE: PATENT AS A SPECIFICATION

From Patent to SRS: Epic to Test Case

A patent and your SRS follow the same four-layer hierarchy, just different names.

1 EPIC	**How can accidents at stop-sign intersections be reduced to save lives and lower costs?**
2 REQUIREMENT	**As a** driver**, I want** to be warned that a stop sign is approaching **so that** I can slow down even if I cannot see the sign.
3 SPECIFICATION	**As a** driver**, I want** A W3-1 warning sign to be installed not less than 100 feet upstream, mounted at 7 feet, visible from 500 feet under daylight conditions **so that** I can slow down even if I cannot see the sign.
4 TEST CASE	Approach at the posted speed. Confirm the warning sign is visible at or before 500 feet. Measure mount height. Pass if both criteria are met with zero deviation else fail.

Working Backwards: Problem to User Story

The patent defines what must be built. Working backwards gives us the problem statement, the Epic, and then the Requirements.

USER / SPONSOR Problem Statements

1. How can accidents at stop-sign intersections be reduced to save lives and lower costs?
2. How can a driver be warned of an upcoming stop sign even if they cannot see it?

Resulting User Stories

As a country, I want fewer stop-sign accidents so that lives are saved.

As a driver, I want to know a stop sign is coming so I can slow down in time.

Review EPICS

The Business Problem

From Problem to Product

Each layer answers a different engineering question.



1	2	3	4	5	6	7	8
Problem Why? A User Need	**Epic** Business outcome	**Feature** Capability	**Business Req** Terms	**Engineering Req** Detailed Terms	**Spec** How Tools	**Design** Details	**Test** Proof

Traceability is the chain: Problem → Epic → Requirement → Specification → Design → Test ➔ FINAL PRODUCT !

An Epic Is the Business Problem

For this course: ONE (1) epic defines the problem the entire team is solving for the semester. Usually stated in the form of a question?

BUSINESS OUTCOME

Describe the measurable result the sponsor needs.

ONE EPIC

Do not create an epic for every feature. The epic is the umbrella. In the form of a question?

BOUNDARY

The epic establishes scope. Features live underneath it.

If a proposed feature does not support the epic, ask why it belongs in the project.

Review Features

The First Level of Understanding the Problem

REVIEW FEATURES

What Is a Feature?

A product capability achieving part of the Epic. Multi-sprint. User-perspective.

S Size & Scope

Spans multiple sprints. Too big for one week, achievable within the semester. Typically, 2-6 stories. Normal completed by 1 person, in this class might be more.

U User-Centered

Framed around a user role and goal, not a technology, database, or endpoint.

T Testable Outcome

Done when all acceptance criteria pass and the team lead signs off, **NOT** when code compiles.

Features Turn the Epic Into Capabilities

1. A feature is larger than a weekly story, usually 3 or more stories and represents something the product can do.
2. A feature has these characteristics

CAPABILITY

A meaningful product capability that delivers part of the epic.

MULTIPLE STORIES

A feature normally requires several stories to complete.

SCOPE CHECK

If a feature cannot be connected to the epic, challenge it.

Epic = outcome. Feature = capability. Story = implementable slice.

REVIEW FEATURES

Features vs. Stories: Where the Line Is

FEATURE

User Registration

Spans multiple sprints. Everything a user needs to create and manage an account.

- Story 1.1: Register with email and password
- Story 1.2: Verify email address
- Story 1.3: Reset forgotten password
- Story 1.4: Update profile information

FEATURE RULES

1. One Feature, General Rule, one owner accountable for every story inside it. Hard for a 15-week course.
2. Estimated in stories, not hours. No stories = not scoped.
3. Done when every story passes acceptance, not when code compiles.
4. AI can draft feature descriptions. The team owns the acceptance criteria.

Review Business Requirements

These are what the Business Wants

What the Business Wants

Business requirements describe outcomes and behavior in customer language.

CUSTOMER LANGUAGE

Avoid database, framework, API, and implementation terms unless the customer requires them.

MEASURABLE

Use quantities, limits, conditions, roles, and observable outcomes. Helps when drafting Specs

PRIORITIZED

Identify what matters most. Not every requirement has equal business value. Never 2 of anything!

The business defines WHAT it needs. The engineering team determines HOW to build it. These raw statements are NOT in the SRS!

The Complete Engineering Specification

This is what the Engineer Produces

How to get to a Specification

Business Requirements vs. Engineering Specifications

BUSINESS REQUIREMENT	ENGINEERING SPECIFICATION
The system shall allow users to log in.	Login via OAuth 2.0. Response under 200 ms at the 95th percentile.
Users shall be able to upload files.	Accept PDF and DOCX up to 25 MB, validated server-side before storage.
The system shall generate project reports.	PDF report in under 3 seconds for datasets up to 10,000 rows.
All user data shall be protected.	Passwords hashed with bcrypt (cost 12). All data in transit uses TLS 1.3.

Requirements Can Have "Levels"

SRS Versions

V1	V2 - V3	V3 - V4
L1 - High *"As a user, I want to post pictures so that I can share them with my friends."*	**L2 - Direct** " As a normal user, I want to be able to add, delete, and update pictures which are saved to a user specific location so that I can share them with my friends	**L3 - Detail** " As a normal user, I want to be able to sort my pictures alphabetically so that I can group them into small categories.
	L2 - Direct " As a normal user, I want to be able to sort my pictures so I can group them into categories.	" As a normal user, I want to be able to sort my pictures by date so that I can group them into small categories.
		" As a normal user, I want to be able to sort my pictures people in the pictures so that I can group them into small categories.

REQUIREMENT LEVELS

L1 → L2 → L3

Decompose large requirements until each piece is small enough to build and test.

L1: Highest Level

The high-level business requirement.
Example: Track employee project time.

L2: More Detail

A major functional capability.
Example: Enter and review project time based on Role.

L3: Story Level

A specific, testable requirement.
Example: Enter time in 0.25-hour increments, ensure compatibility.

This is a process, not a 1 and done. Each person should have 1 L3 story / week

Every lower level must trace upward to a valid business need.

What Is a Specification?

A specification defines how a requirement will be built. Precise enough that a skilled practitioner needs no further guidance.

R Requirement

The WHAT

Stated by the customer. Engineers fulfill requirements, not write them.

S Specification

The HOW

Written by the engineer. Numbered, Priority order, measurable, and uniquely identified.

A Artifact

The BUILD

The deliverable produced by following the specification exactly. Testable.

L Skilled

Legal Standard

If a spec requires guessing to implement, it is legally and technically inadequate.

Turning Business Speak into Engineering Terms

A specification removes ambiguity so a skilled practitioner can build the artifact.

INPUT

Business Terms, needs, givens, constraints, assumptions

NOT IN SRS

PROCESS (A)

Create the Engineering User Stories or Engineering Requirements

SRS

PROCESS (B)

Break the User Stories into specifications 1:1 or 1:Many

Noting: Assumptions & Constraints SRS

OUTPUT

Engineering Specifications. Must be signed off by the business

SRS

A good specification answers the builder's questions before the builder has to ask them.

ENGINEERING ROLES

Who Produces Specifications?

EA Enterprise Architect

Defines the problem space before any spec is written. Works with compliance, security, and leadership across multiple epics.

SA Solutions Architect

Translates the EA specification into an executable design. Typically owns one or more epics. Uses AI agents to accelerate design drafts.

AI GenAI Engineer

Builds on large language models. Often spec writer and builder in one. Must disclose AI use in every artifact.

ML MLOps Specialist

Deploys and monitors ML systems. Owns the operational spec: SLAs, drift detection, retraining triggers.

AO AI Orchestration

Designs multi-agent systems. The spec defines each agent's scope, handoff format, and failure mode.

SPECIFICATION COMPONENTS

What Goes Into a Specification?

Every measurable part of an artifact belongs in the spec, enough that a skilled practitioner can build it without asking a single question. *(can AI create each of these?)*

I I/ O Format

User screen mockups, API schemas, and data field definitions.

A Algorithm

The logic converting input to output, shown via a flowchart.

D DB Schema

Table definitions, field types, indexes, and relationships.

C Class Diagrams

Object relationships and interaction flows for software-intensive systems.

B Block Diagrams

System-level component relationships for hardware and embedded projects.

3 3D Views / Schematics

Physical dimensions, circuit schematics, and layout drawings for hardware.

YOUR FIRST DELIVERABLE

SRS Contents and Key Rules

S	**One Spec per Requirement**	Every requirement needs at least one specification; it is not “wrong” to write more than 1.
T	**One Test Case per Spec**	Every specification needs at least one test case, with special attention to edge case(s). *Details comes latter!*
A	**Acceptance Criteria**	Every specification must define exactly what passing looks like; the contract with the sponsor.
D	**Define All Terms**	Every business and technology term must appear in the glossary, assume nothing.
C	**Change Log Required**	Every requirement and specification must have a log tracking who changed it and when.

REQUIRED DEFINITIONS

Assumptions and Constraints

IT Assumption

A statement accepted as true for planning, even though it has not been proven.

If an assumption proves false it introduces risk: delays, redesign, or project failure.

Example: Email addresses follow the name@domain format.

IT Constraint

A limitation that restricts the possible solutions or design of a system.

Constraints may be technical, such as software limits, or business-driven, such as budget or regulatory requirements.

Example: Only Google Chrome is supported.

WORKED EXAMPLE — PART 1 OF 2

Full Specification: Graphical Login Screen

User Story: As a user, I want a graphical login screen so I can create an account and return to it without re-registering.

Specification Details

As an Admin, I want to allow users to create a new account with email and password or log in to an existing account so they can do their work.

Constraints

- No email address may be duplicated.
- Passwords must be masked
- Passwords: 8 characters minimum
- 1 upper case letter, 1 number

Assumptions

Email addresses must follow the name@domain format

Additional Specification

The login page must use high-contrast colors on a clean background.
The login form displays only: User ID, Password, and a legal disclaimer stating unauthorized use is prohibited.

WORKED EXAMPLE

Test Cases, Acceptance Criteria, Change Log

Test Cases

Test 1 — Pass

- Open Chrome and navigate to https://xyz.com.
- Enter a valid email and a compliant password.
- Confirm login, log out, then log in again successfully.

Test 2 — Failure

Attempt to register a duplicate email. Verify the system rejects it without creating a duplicate account.

Acceptance Criteria

Tests 1 and 2 complete with no user-facing errors and no system log errors, reproducible by an independent reviewer.

Change Log

Track every change after the first graded submission: Who & What & why!

Sept 20: James: Added password special-character complexity rule. Req 1.2

Oct 30: Carly: Expanded accepted email domains to include .edu. Req 3.1

Nov 14: Kato: Added legal disclaimer text requirement to the login form. Req 4.2

The End-2-End Process

What does the solution look like?

BREAKING DOWN THE PROBLEM

Epics → Features → Stories (Requirements)

Epic (1 for this course)

A major business problem or user need, often cross-cutting several functions.

→

Features (multiples)

A specific product capability that achieves part of the epic. (Takes more than 1 week)

→

Story (1/pers/week)

A concise piece of functionality from the user's perspective, completable in under a week.

WORKED EXAMPLE: THE RIVER CROSSING

EPIC

How can I move 100 people/day, carrying up to 100 lbs, across this river at under 1-in-1,000,000 risk of falling in?

→

FEATURE

A pedestrian crossing capable of 100 person-trips per day in all weather conditions.

→

L1: STORY

A single-span footbridge with a load-rated, non-slip walking surface and guard rails on both sides.

Requirement: As a city engineer, **I want** a load-rated footbridge with non-slip decking and rails, **so that** 100 people a day can cross safely in any weather.

Acceptance criterion: bridge passes load testing at 100 simultaneous 100 lb crossings with zero structural deflection beyond code.

EXAMPLE ONLY!

Time Tracking System - Epic

The user requires our team to develop a complete, web-based time tracking system that enables an organization to define and manage projects, assign people to projects, capture individual time in 15-minute increments, and produce accurate project and person-based reporting for project management, resource utilization, cost tracking, and accountability.

Major Features

1. *Project & People Management:* create projects, maintain supporting details, and manage assignments.
2. *Time Entry & Tracking:* record individual time in 15-minute increments and calculate totals/cost.
3. *Reporting & Analysis:* provide project-based and person-based reporting for management.

Security Requirements

1. Administrator / Project Manager: manages projects, people, assignments, and project information.
2. Team Member / Employee: records and maintains time against assigned projects.
3. Report Viewer / Management: reviews project and individual time/cost reports.

EXAMPLE ONLY!

Project Management and Time Tracking – Features & Stories

1. Project & People Management

Establish the projects, people, and relationships required for time tracking.

Story 1.1 — Create Project

As a Project Manager, I want to create a project and enter its supporting details, so that the organization has a centralized project record.

Project name, description, client/customer, manager, dates, status, team, budgeted hours, and project cost/budget.

Story 1.2 — Assign People

As a Project Manager, I want to assign people to a project, so that team members can record time against authorized projects.

2. Time Entry & Tracking

Capture accurate individual time against assigned projects.

Story 2.1 — Enter Time

As a Team Member, I want to enter time worked on an assigned project in 15-minute increments, so that my work is accurately recorded.

Users enter 0.25-hour increments with date and optional activity/description.

Story 2.2 — Review Time

Users can review and correct previously entered time, subject to system rules.

Story 2.3 — Calculate Project Cost

As a Project Manager, I want the system to calculate cost from recorded hours and applicable hourly rate.

3. Reporting & Analysis

Turn captured time into useful management information.

Story 3.1 — Project-Based Report

As a Project Manager, I want to generate a report by project, so that I can understand time and cost spent on each project.

Include project, status, team, total hours, hours by person, project cost, budget, and variance.

Story 3.2 — Person-Based Report

As a Manager, I want a report by person, so that I can understand how an individual's time is allocated across projects.

Include person, projects, hours, total hours, cost, and date/time period.

Story 3.3 — Filter Reports

As a Report Viewer, I want to filter by project, person, status, and date range.

EXAMPLE ONLY!

Project Management and Time Tracking – Specification Examples

Story 2.1 — Enter Time

As a Team Member, I want to enter time worked on an assigned project in 15-minute increments, so that my work is accurately recorded.

Specification 2.1.1

Select Assigned Project and Date

As a team member, I want to select an assigned project and the date I worked on it, so that my time is recorded against the correct project and work date.

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EXAMPLE ONLY!

Assumptions, Constraints & Test Cases (Sample)

Assumptions

a) **Users are assigned to projects** before they can enter time against those projects.
b) **Time is recorded in hours using 15-minute increments**, with 0.25 hours representing 15 minutes.

Constraints

a) **Time entries must be entered in 15-minute increments**; values such as 0.10 or 0.30 hours are not permitted.
b) **Team members can only enter time for projects to which they are assigned.**

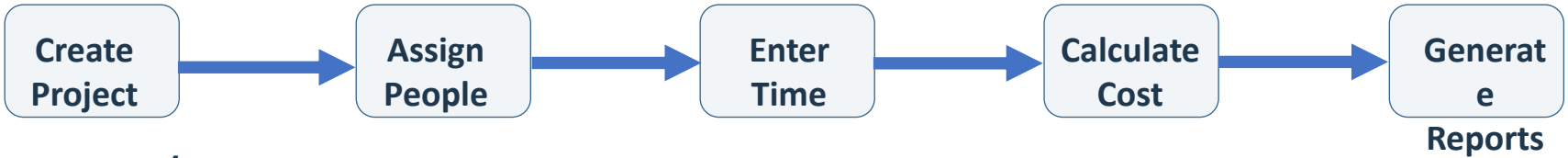
Test Cases

Test Case	Test Description	Expected Result
TC-2.1.2-01	Enter 0.25 hours for an assigned project.	System accepts and saves the 0.25-hour time entry.
TC-2.1.2-02	Enter 1.50 hours for an assigned project.	System accepts and saves the 1.50-hour time entry.
TC-2.1.2-03	Enter 0.30 hours for an assigned project.	System rejects the entry and informs the user that time must be entered in 15-minute (0.25-hour) increments.

EXAMPLE ONLY!

Definition of Done

EXPECTED END-TO-END WORKFLOW



Project / Epic SUCCESS CRITERIA

- Demonstrate a working end-to-end system, not simply individual screens or database components.
- Data entered into projects, people, and time tracking must flow correctly through the application.
- Project and person reports must accurately reflect recorded hours and associated costs.
- The team should be able to demonstrate the complete workflow during the final project presentation.
- Supporting technical work (authentication, Git integration, attachments, deployment, etc.) should support the core business features rather than expand the business scope.

End

Week 4 — Requirements, Specifications & Test Plans